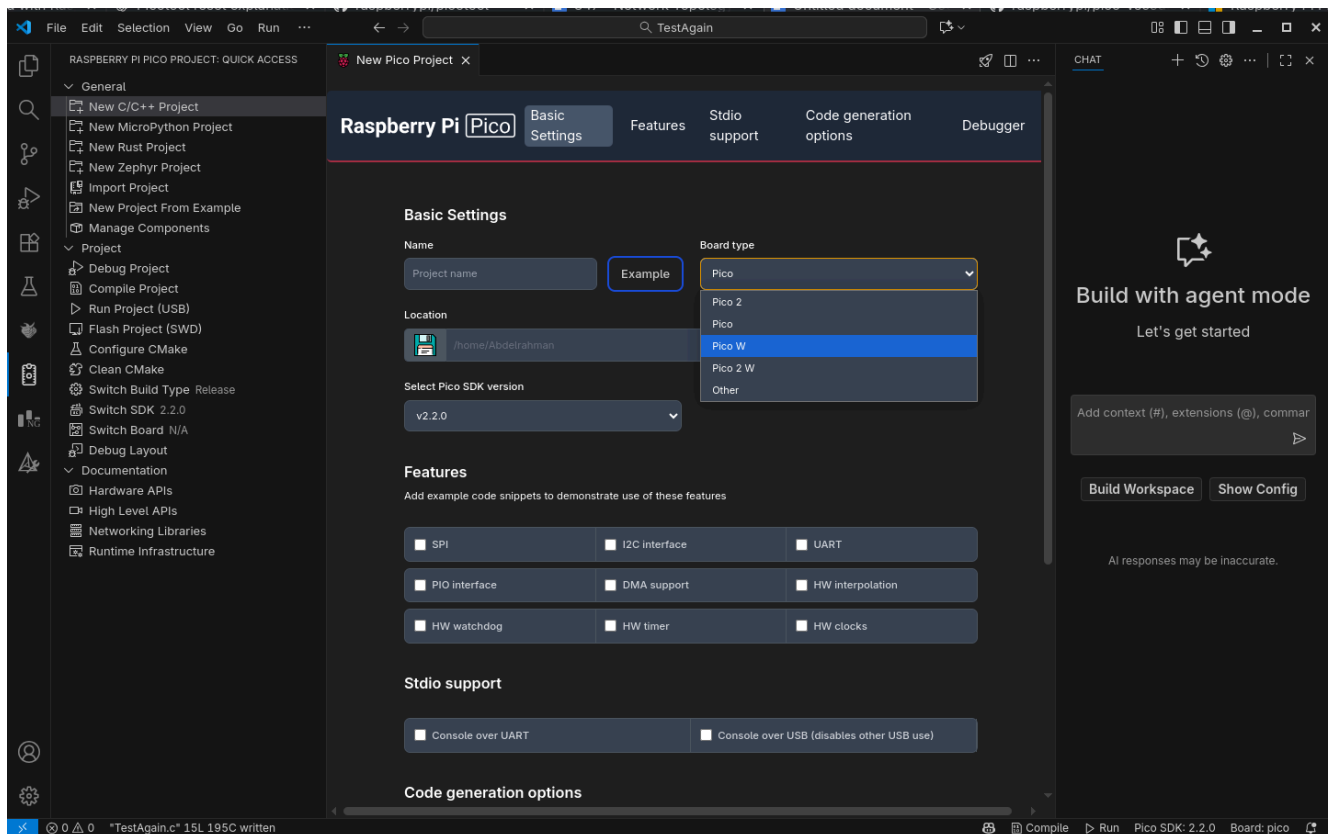
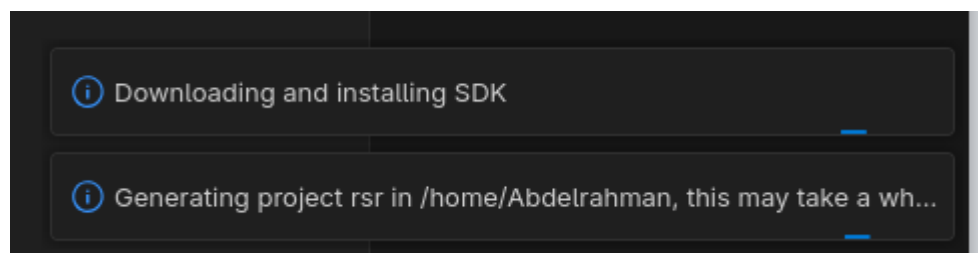


Using the VS-Code extension

For anyone having issues with the manual setup, install the visual studio code extension from [this link](#). After installing the extension, create a new C/C++ project



Give the project a name and select the correct board type, then scroll down and select create



After that it will start to install the SDK, the toolchain and all of the needed utilities



After the project is created, two options should appear below, Compile and Run. Compile only creates the uf2 file but you have to move it manually to the RP2040 drive. The Run option flashes the program for you. According to the datasheet the run option can run even if the RP2040 is not in BOOTSEL mode, but this is not guaranteed.

FAQ

1. The microcontroller won't appear again after I flashed the uf2 file the first time

Make sure when you are connecting the microcontroller to your laptop you are pressing the BOOTSEL button **before and during** the connection. The BOOTSEL button is just connecting an internal pin to GND, and this pin must be grounded during the boot so the pi pico boots into USB flash mode.

Other thing you can try is to use picotool.exe, you will find it in
C:/users/[your-user]/.pico-sdk/picotool/[version]/picotool/

And use the following command

picotool.exe load [path to your uf2] -f

2. Is my LED active high/low ?

The logic type is a thing for input pins, LEDs are output peripherals, they have two legs, the longer one is the anode and the shorter is the cathode. You control the LED by connecting the cathode with the ground and connect the anode with a GPIO pin that you control, the LED will light when you put HIGH voltage on the pin.

RGB LEDs are different, they have 4 legs, the longest one is a common cathode or a common anode (the seller should tell you). If it is a common anode you connect the longer leg to VCC and control it by putting LOW voltage on the other legs using GPIO pins, and the opposite is done for common cathode LED (connect the longest leg to GND)

3. Do I need resistors with the LEDs ? It is working without the resistors

Yes it will work, but it will burn out quicker, so please use resistors, also make sure you aren't using high resistance ones (use less than 330 ohm resistors)

4. There are errors in the includes

Ignore it for now, if it compiles then it is fine, there is a conflict between VScode's language diagnostics and the CMake extension. If you get asked to select a kit, ignore it and most probably you won't get include errors. We will look for a better fix later

Further Reading

1. <https://datasheets.raspberrypi.com/pico/getting-started-with-pico.pdf> - Getting started guide and more details on the extension and how to use another pico for rescuing a broken one
2. <https://datasheets.raspberrypi.com/rp2040/rp2040-datasheet.pdf> - RP2040 datasheet
3. We recommend going through the milestone PDF and giving yourself time to understand what each step is doing, why do we need a cross-compiler ? What is the build system ? etc..